

SESSION 3

Securities Trading & Order Execution

Why good investment ideas fail because of poor trade execution

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What Happens Between Idea and Profit? The Execution Gap



Good
Analysis



Correct
Investment



Poor Order
Execution



Disappointing
Return

What courses teach

Finance courses focus on WHAT to buy — valuation, industry analysis, earnings forecasts, DCF models, and ratios.

What professionals do

Professionals obsess over HOW to buy — order type, timing, liquidity, cost minimisation, execution benchmarks.

The costly gap

A correct stock pick + poor execution can deliver worse returns than an average pick + excellent execution.



Session Theme: Investment returns are determined not only by stock selection — but by execution quality.

Evolution of Indian Trading: From Ring to Algorithm

Open Outcry

Pre-1994

BSE's physical trading ring. Jobbers and brokers traded face-to-face. Information asymmetry was extreme; prices varied by trader connections.

📌 BSE founded 1875 — Asia's oldest exchange. Ring system favoured insiders; retail investors had almost no price transparency.

 bseindia.com/about

BOLT + NEAT

1994–2000

NSE launched NEAT — India's first fully electronic trading system. BSE responded with BOLT. Screen-based trading replaced the ring entirely.

📌 NSE's 1994 VSAT network connected 400+ cities. By 2000, NSE surpassed BSE in equity turnover — technology created a new market leader.

 nseindia.com/history

Online Brokers

2000–2015

Internet trading democratised access. ICICI Direct (2000) was India's first internet platform. Zero-brokerage Zerodha (2010) disrupted the model entirely.

📌 Zerodha: 0 → 7 crore+ clients by 2024 with flat ₹20/trade. Disrupted ₹5,000 cr brokerage industry. Now handles 15%+ of NSE volumes.

 zerodha.com/story

Algo + AI

2015–Present

Co-location at NSE allows HFT firms to execute in microseconds. SEBI permits algo trading via API. AI reads filings, news, and social sentiment.

📌 Over 50% of NSE volume is now algorithmic. Renaissance Medallion Fund: ~66% gross annual return over 30 years using pure quant execution.

 sebi.gov.in/algo-trading

Has Technology Improved Investing — or Just Made Markets Faster?

✓ BENEFITS OF TECHNOLOGY

Price Transparency

Real-time quotes eliminate information monopolies. Any retail investor sees the same Level 2 order book as a hedge fund.

Lower Costs

Zero-brokerage models: Zerodha ₹20/trade vs traditional ₹500+. Costs that once erased small returns now barely register.

24/7 Access

Mobile apps, instant execution, real-time P&L. Commodity & currency markets trade 23+ hours. Access is broader than ever.

Liquidity Depth

Algo market-makers provide continuous bid-ask quotes. Spreads on Nifty 50 stocks are fractions of a paise vs rupee-wide pre-2000.

⚠ RISKS CREATED BY TECHNOLOGY

Flash Crashes

US Flash Crash (May 2010): \$1 trillion erased in 36 minutes by algo momentum. India's Nifty saw 900-point intraday swings on algo triggers.

Overtrading

Easy access = more trades. SEBI 2023 study: 89% of F&O traders lose money. Low friction removes natural deterrents to speculation.

Herding & Panic

Algos trained on similar data react identically, amplifying crashes. COVID March 2020: Nifty fell 38% in 40 days as algos and retail panic-sold.

Info Overload

Noise exceeds signal. 24-hr news, social media, WhatsApp tips. More data does not equal better decisions — often the reverse.

Order Types: How Your Order Choice Determines Your Return

⚡ Market Order

Use: Immediate fill at best available price

- ✓ Guaranteed execution in liquid stocks
- ✗ Price uncertainty — you receive the ask, not the last trade

📌 Retail trader buys Adani Enterprises in a surge: ₹2,850 market order filled at ₹2,910 —

nseindia.com/products/equities-market-orders

🎯 Limit Order

Use: Execute only at your specified price or better

- ✓ Full price control — never overpay
- ✗ Non-execution risk if stock moves away

📌 Buffett's Berkshire: set a maximum price, wait patiently. Missed Walmart early 1990s by

berkshirehathaway.com/letters

🛡️ Stop-Loss Order

Use: Auto-exit when price drops below threshold

- ✓ Prevents catastrophic losses; enforces discipline
- ✗ False triggers; gap-downs bypass stop entirely

📌 COVID crash Mar 2020: Nifty fell 13% in a single day. Stop-losses at -5% were bypassed —

nseindia.com/market-data

🧊 Iceberg Order

Use: Show only a fraction of total size to the market

- ✓ Hides institutional demand; minimises market impact
- ✗ Partial fills; sophisticated algos can detect the pattern

📌 EPFO's quarterly ETF buy (₹5,000–7,000 cr/quarter): sliced into weeks of small orders vi

epfindia.gov.in/annual-report

🕒 GTT Order

Use: Good-Till-Triggered: fires at a future price

- ✓ Set-and-forget discipline; no daily monitoring
- ✗ Market conditions may shift before trigger fires

📌 Long-term investor sets GTT buy at ₹3,000 on TCS (current ₹3,500). Order sits dormant until it reaches 3,000

zerodha.com/gtt

📐 Bracket / Cover

Use: Entry + stop-loss + target automated in one order

- ✓ Enforces R:R discipline automatically
- ✗ Intraday only; false triggers can exit prematurely

📌 Intraday Bank Nifty: Buy 44,500, stop 44,300, target 44,900. Risk ₹200, reward ₹400 (1:2)

nseindia.com/trading-products

Stop Losses: Capital Protection Tool or Forced Mistake?

✓ CASE FOR STOP LOSSES

Enforce discipline

Remove emotion from exit decisions. Pre-committed stops cannot be overridden by hope or denial. Critical for traders.

Limit catastrophic loss

-10% → -20% → -40%: each level exponentially harder to recover. Stop losses contain the damage before it compounds.

Capital preservation

Rule #1: don't lose money. Stop losses ensure survival to participate in the next opportunity.

Industry standard

Every professional trader pre-defines maximum loss per trade. Risk management begins by defining the loss, not the profit.

⚠ CASE AGAINST (LONG-TERM INVESTORS)

False triggers

Intraday volatility repeatedly hits stops in fundamentally healthy stocks — exit forces, stock recovers, you sold the bottom.

Gap-down bypass

Stops don't prevent gap losses. Nifty opened -6% on COVID Mar 23. A stop at -3% executed far below your target price.

Buffett's alternative

Position sizing over stop losses. If a position is small enough, temporary volatility is tolerable. 'Time is friend of good business.'

Locks in permanent loss

Selling sound companies during panics converts a temporary mark-to-market loss into a permanent capital loss.



How Institutions Trade: The Invisible Infrastructure

Mutual Funds (AMCs)

- ▶ Split large orders over multiple days
- ▶ Track VWAP as execution benchmark
- ▶ Avoid signalling trades to market
- ▶ Use basket orders for rebalancing

 SBI MF deploying ₹1,000 cr into Infosys: splits into 50–100 orders over 3–5 days. A single block trade would push price 2–3%, costing ₹20–30 cr in impact cost.

 amfiindia.com/research-insights

Pension Funds (EPFO)

- ▶ Prioritise liquidity above all else
- ▶ ETF-only equity exposure mandate
- ▶ Long-term, low-turnover approach
- ▶ Benchmark to index, not active returns

 EPFO deploys ₹5,000–7,000 crore quarterly into Nifty 50 and Nifty Next 50 ETFs. T+1 settlement requirement — ETFs are the only viable execution vehicle.

 epfindia.gov.in/annual-report

LIC (Insurance)

- ▶ Countercyclical buyer during crashes
- ▶ Manages ₹59.17 lakh crore AUM
- ▶ Acts as market stabiliser in crises
- ▶ Long-duration liability match required

 COVID crash Mar 2020: LIC deployed ₹10,000+ crore as Sensex fell from 41,000 to 25,800. This institutional buying provided crucial market support.

Hedge Funds / HFTs

- ▶ Co-location at NSE for microsecond speed
- ▶ Algorithmic execution — no manual trading
- ▶ Minimise information leakage at all costs
- ▶ Statistical arbitrage + momentum models

 Renaissance Medallion Fund: ~66% gross annual return over 30 years (1988–2018). Zero human stock picks — pure mathematical pattern execution.

 sec.gov/cgi-bin/browse-edgar/renaissance



Key Question: Why do institutions spend millions on trading infrastructure when brokerage is near-zero?

VWAP: The Metric Retail Investors Ignore, Professionals Live By

VWAP = Volume-Weighted Average Price

$\text{Sum}(\text{Price} \times \text{Volume}) \div \text{Total Volume}$ | Calculated continuously throughout the trading day

What VWAP Measures

The true average price paid for a stock during the day, weighted by volume. Heavy trading at higher prices pulls VWAP up — and vice versa.

Why Institutions Care

Fund managers are evaluated on execution quality — not just returns. A fund that consistently buys above VWAP has a structural cost disadvantage.

VWAP as KPI

Desk heads at mutual funds have daily VWAP compliance reports. Buying below VWAP = excellent execution. Above VWAP = poor execution.

Limitations

VWAP is backward-looking — it tells you what the average price WAS. Gaming VWAP (bunching orders near day end) is a known institutional strategy.



Circuit Breakers and Market Panic: Control or Delay?

SEBI Circuit Breaker Framework — "Market-Wide Circuit Breakers"

10% Move

45-min halt (before 1pm) | 15-min halt (1pm–2:30pm) | No halt after 2:30pm

15% Move

1hr 45min halt (before 1pm) | 45-min halt (1pm–2pm) | Rest-of-day halt after 2pm

20% Move

Trading halted for the REMAINDER of the day — no exceptions

CASE STUDY: COVID Circuit Breakers — India (February–March 2020)

Feb 28

Nifty
-3.8%

Mar 9

Nifty
-5.5%

Mar 13

Nifty
+4.4%

Mar 16

Nifty
-7.9%

Mar 23

Nifty
-13%

Apr 3

Recovery
begins

Case FOR Circuit Breakers

Pause panic selling. Allow information to be processed. Prevent cascading margin calls. Protect systemic stability.

Case AGAINST Circuit Breakers

Delay price discovery. Panic resumes after reopening. Creates artificial price discontinuity. Cures the symptom not the cause.

The Hidden Cost of Trading: What Retail Investors Ignore

COST WATERFALL: ₹1,00,000 EQUITY DELIVERY TRADE (BUY + SELL ROUND-TRIP)

Brokerage	₹0 – ₹20/trade	(0.01–0.05%)	Zerodha flat fee
STT (buy)	0.1% delivery	(0.1%)	SEBI-mandated
Exchange Charges	~0.00335%	(~0.003%)	NSE + BSE fee
Stamp Duty	0.015% equity	(0.015%)	State government
Slippage	0.1–0.5%+	(0.1–3%)	Market order gap
Impact Cost	0.5–2%+ large	(0.5–5%)	MF / institutional

Total round-trip (delivery): 0.3–0.4% explicit + 0.1–3% slippage + 0–5%+ impact cost (for institutional sizes)

🔑 Hidden costs (slippage + impact) often exceed visible brokerage 10x for large trades. Institutions care about VWAP more than commission.

What Theory Teach vs What Professionals Actually Do

Concept	Textbook Says	Professionals Do	Why It Differs
Order Execution	Use market orders when speed matters	Rarely use large market orders — always limit or algo	Impact cost can exceed brokerage 10x at institutional sizes
Stop Losses	Always use stop losses to manage risk	Position sizing first; stops only for active trading book	Good businesses have temporary volatility — don't get shaken
VWAP	Historical execution benchmark metric	Daily KPI reported on every institutional trading desk	Institutions judged on execution quality, not just returns
Trading Costs	Brokerage is the main transaction cost	Focus on slippage and impact — brokerage is trivial	Hidden costs dominate explicit costs for any real trade size
Margins	Leverage enhances expected returns	Use selectively; risk management beats return maximization	Margin calls in crisis = forced selling at worst possible time
Liquidity	Liquidity enables easier trading	Analyse depth before entering; exiting is the true skill	You can always enter. Exiting without price impact is hard.



The gap between textbook execution theory and professional practice is where most retail trading losses are generated.

Six Case Studies: Execution, Discipline & Market Behaviour

Warren Buffett

Trading Discipline

Berkshire Hathaway rarely trades. 'Our favourite holding period is forever.' ~19.8% CAGR 1965–2023 — achieved by NOT trading, not by trading cleverly.

 berkshirehathaway.com/letters

Renaissance Tech

Quantitative Edge

Medallion Fund: ~66% gross return annually over 30 years. Pure algo execution, zero human stock-picking. Speed + math consistently outperforms intuition at scale.

 sec.gov/cgi-bin/browse-edgar/renaissance

Vanguard

Cost Minimization

John Bogle: costs compound just like returns — against you. Vanguard S&P 500 TER: 0.03% vs active average 1%+. The difference compounds to millions over decades.

 investor.vanguard.com/investment-products

BlackRock Aladdin

Institutional Execution

\$10 trillion AUM. Aladdin processes 250,000 trade scenarios per week. Risk management and execution infrastructure is the competitive moat — not stock-picking.

 blackrock.com/aladdin

COVID Crash 2020

Panic & Circuit Breakers

Nifty: 12,430 → 7,610 in 40 days. Circuit breakers triggered multiple times. Retail sold; LIC/institutions bought. Nifty recovered 80%+ within 6 months.

 nseindia.com/market-data/historical-data

2008 Crisis

Liquidity Collapse

Lehman collapse: \$600B in assets, zero liquidity in 24 hours. 'Liquidity is abundant until everyone needs it simultaneously.' — the classic systemic risk lesson.

 federalreserve.gov/2008-crisis-report

The Future of Trading: Will Human Traders Become Obsolete?

WHAT HUMANS DO BETTER

Qualitative judgment

Regulatory changes, geopolitical shifts, management character — context humans grasp that historical models miss entirely.

Regime change detection

2008, COVID, Russia-Ukraine — fundamental regime changes require human judgment. Algos trained on history can't price unprecedented events.

Client relationships

Institutional PMs manage client trust, not just portfolios. Clients want human accountability and explanations during crises.

Value philosophy

Buffett/Munger: understand the business deeply, hold forever. This level of qualitative judgment is beyond current AI capabilities.

WHAT AI / ALGO DOES BETTER

Speed & scale

AI analyses 10,000 earnings reports, social sentiment, and satellite imagery simultaneously — no human can match this breadth or speed.

Execution excellence

Algo execution eliminates emotion, optimises VWAP, manages order flow at microsecond level. No fatigue, no fear, no greed.

Pattern recognition

ML identifies statistical patterns across decades of price history invisible to human eyes — the Renaissance Technologies advantage.

Risk monitoring

BlackRock's Aladdin monitors millions of positions across asset classes 24/7 in real-time — impossible to replicate manually.



The future is augmented intelligence — human judgment + AI execution + data science = next-generation professional investing.

Likely Questions & Responses

Q

If markets are efficient, why do professional investors exist?

► Markets are reasonably efficient most of the time — but not perfectly efficient all the time. Professionals compete to find temporary inefficiencies, execute better, and manage the behavioural biases retail investors cannot control.

Q

Why not always use market orders for simplicity?

► Execution certainty comes at the cost of price uncertainty. For small retail trades in liquid stocks, fine. For any meaningful institutional size, market order impact cost can dwarf brokerage many times over.

Q

Can AI eventually replace fundamental analysts?

► AI will replace routine screening and pattern recognition. It will not replace judgment on regime changes, new industries, qualitative management assessment, or geopolitical risks — yet. The real threat is AI augmenting analysts, making each human 10x more productive.

Q

What is the most underestimated investment risk?

► Liquidity risk. You can always enter a position. Exiting a large holding in a thinly-traded stock without destroying the price takes skill, patience, and technique. The 2008 crisis was fundamentally a liquidity crisis, not a fundamental collapse.

Q

Can low-cost passive investing outperform active management?

► Yes, consistently over long horizons. SPIVA India 2023: 73%+ of active large-cap funds underperformed Nifty 100 over 5 years after fees. Costs compound against you exactly as returns compound for you. Source: spiva.spglobal.com

SESSION RECAP

Three Big Ideas

01

Most investors focus on WHAT to buy; professionals obsess over HOW to buy

Order type, timing, liquidity, execution cost, VWAP benchmarking — these are the tools of professional investing. Ignoring them turns correct analysis into poor returns.

02

Technology transformed markets — but has not eliminated human emotion

Speed, transparency, and lower costs are real benefits. But algorithms amplify panic, overtrading destroys returns, and 89% of F&O retail traders still lose money in the digital age.

03

Execution quality creates performance differences between identical stock picks

Two investors buy the same stock on the same day. One uses a market order at open; one uses a patient limit order below VWAP. Their returns diverge before the business even performs.



Final Board Question: If Warren Buffett and a novice buy the same stock today — why will their long-term outcomes still be completely different?